

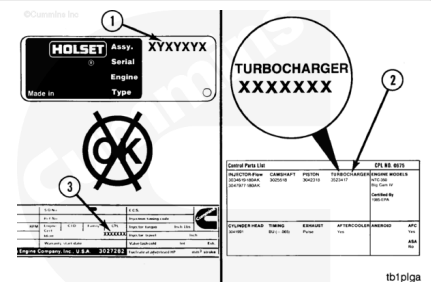
Trainings ▾

Install

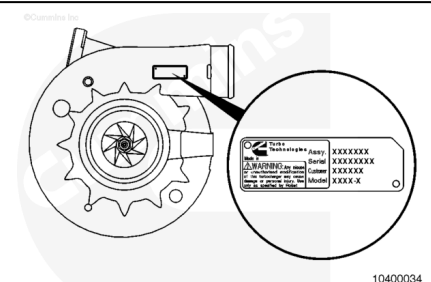
Check for Correct Component

Compare the assembly number (1) on the turbocharger dataplate with the turbocharger specified in the Control Parts List (CPL), Bulletin 3379133 (2).

The CPL number for each engine is listed on the engine's dataplate (3).



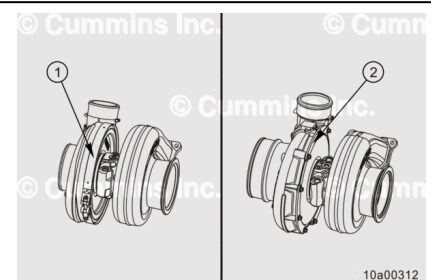
Holset® turbochargers are used on the QSK45 and QSK60 engines. Use the data tag on the turbocharger to determine the model.



The HE850 model name is unable to be retroactively applied to early design releases due to regulatory compliance constraints. These HE850 turbochargers will still have HE800 as the model name listed on the turbocharger dataplate. For differentiation from the standard HE800 turbocharger, the HE850 model name will be referenced throughout this procedure.

The procedure and specifications will be different based on the type of turbocharger. Identify the type of turbocharger equipped:

- The HE800 turbocharger (1) can be identified by the V-band joint between the compressor and bearing housing.
- The HE850 turbocharger (2) can be identified by the bolted joint between the compressor and bearing housing.



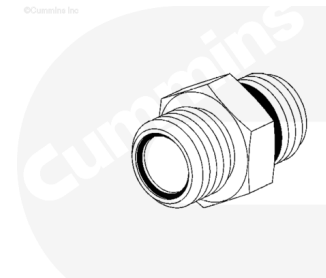
One turbocharger is mounted directly on the exhaust manifold.

Both the HE800 turbocharger and HE850 turbocharger are cooled using coolant from the high-temperature circuit.

Each HE800 turbocharger weighs approximately 57 kg [126 lb]. Each HE850 turbocharger weighs approximately 65 kg [143 lb]. Both will require proper lifting equipment.

Note : All turbocharger oil and water fittings are the flat-face o-ring types of fittings. Make sure the o-ring is in place before attaching the hose(s) to the fittings.

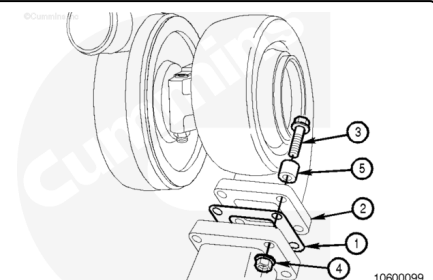
The Holset® turbocharger uses a 9/16-18 UNF, straight-thread, o-ring type of oil supply fitting.



QSK45 Single Stage

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not reuse old capscrews (3) or flange nuts (4). Discard these components. Reusing these components can contribute to exhaust leaks.

Note : Make sure that the turbocharger is correctly aligned during installation to prevent damage to the turbocharger bellows.

Apply an anti-seize compound to the mounting capscrews threads.

The word OUT or the raised gasket bead **must** be positioned toward the turbocharger.

Install the gasket.

Do **not** install the turbocharger without spacers.

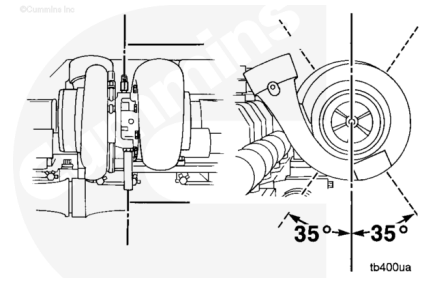
Install the turbocharger (2) and gasket (1) with four capscrews (3), spacers (5), and flange nuts (4).

Tighten the flange nuts (4).

Torque Value: 55 n•m [40 ft-lb]

The turbocharger uses capscrews to attach the bearing housing to the turbine housing.

Position the turbocharger drain tube. The drain tube **must** be within 35 degrees of vertical. Turn the bearing housing to align the tube, if necessary.

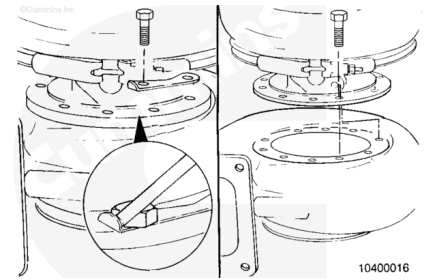


Bearing Housing Adjustment

Bend the lockplate off the capscrew heads.

Remove the capscrews to allow the bearing housing to rotate.

Lift the compressor and bearing housing. Rotate the bearing housing. Align the drain tube and the capscrew holes.



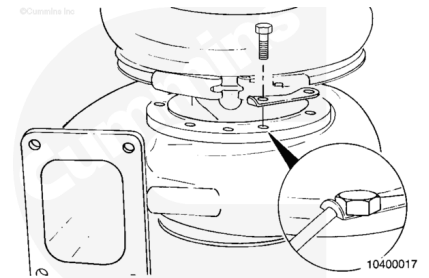
Apply anti-seize compound, Part Number 3824879, to the capscrews.

Install the lockplates and capscrews.

Tighten the capscrews.

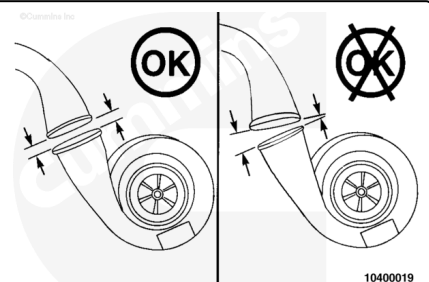
Torque Value: 21 n•m [186 in-lb]

Bend the lockplate tabs over the capscrews.



Check the air crossover-to-turbocharger alignment.

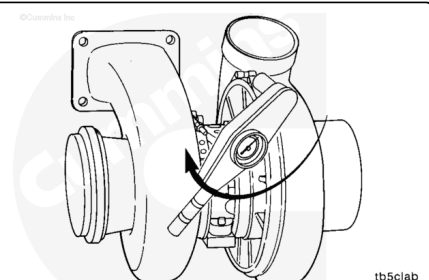
- Install the air crossover on the aftercooler.
- Look at the alignment between the crossover and the turbocharger compressor outlet.
- Remove the crossover and proceed with the assembly if the alignment is correct.
- Proceed to the next step to adjust the compressor housing if the alignment is **not** correct.



To adjust the turbocharger compressor housing alignment:

- Loosen the V-band clamp.
- Turn the compressor housing to the correct alignment.
- Tighten the clamp.

Torque Value: 9 n•m [80 in-lb]

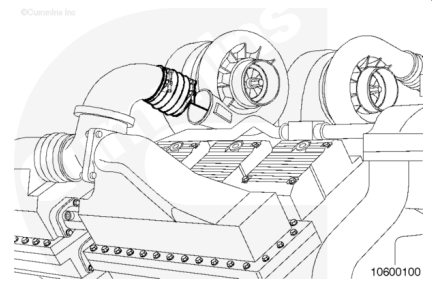


- Tap the clamp with a mallet. Tighten the clamp again.

Install the hose and two hose clamps between the turbocharger and the aftercooler inlet connection.

Tighten the clamps.

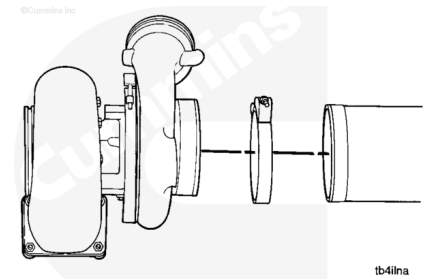
Torque Value: 9 n•m [80 in-lb]



Connect the inlet piping to the turbocharger.

Tighten the clamps.

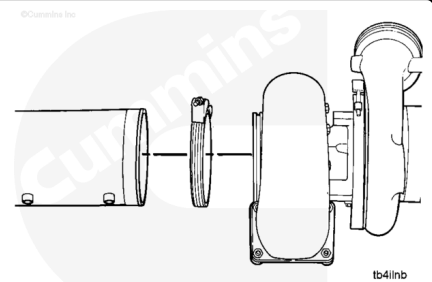
Torque Value: 9 n•m [80 in-lb]



Connect the exhaust piping to the turbocharger.

Tighten the V-band clamp.

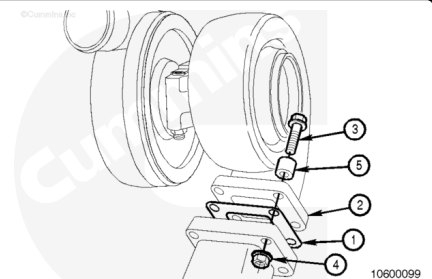
Torque Value: 9 n•m [80 in-lb]



QSK60 Single Stage Industrial and Power Generation

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not reuse old capscrews (3) or flange nuts (4). Discard these components. Reusing these components can contribute to exhaust leaks.

⚠ CAUTION ⚠

Check to be sure the correct turbocharger assembly is installed. The turbocharger assembly number is on the dataplate that is mounted on the turbocharger. Compare the assembly number with the number in the Control Parts List (CPL) manual with the corresponding CPL for the engine being serviced. Many turbochargers appear the same physically, but contain different parts. If the wrong turbocharger assembly is installed, the performance of the engine will be less than specification, or damage can result to the engine due to excessive cylinder pressure and temperature.

Note : Replacement turbochargers are often built with a standard turbine housing and compressor housing orientation. These housing orientations on the replacement turbocharger may not match the housing orientations of the turbocharger that is being replaced. Make sure that the turbocharger is correctly aligned during installation to prevent damage to the turbocharger bellows.

Remove any protective caps and tape used to cover turbine inlet and exhaust manifold outlet.

Apply film of copper-free, high temperature anti-seize compound, Part Number 3824879, or equivalent, to turbocharger mounting capscrews.

The word OUT or the raised gasket bead **must** be positioned toward the turbocharger.

Install the gasket.

Do **not** install the turbocharger without spacers.

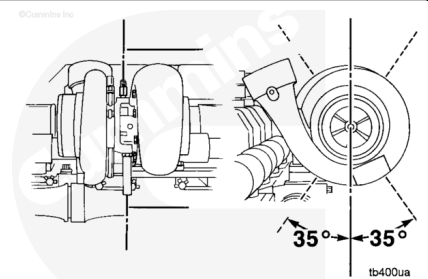
Install the turbocharger (2) and gasket (1) with four capscrews (3), spacers (5), and flange nuts (4).

Tighten the flange nuts (4).

Torque Value: 55 n•m [40 ft-lb]

The turbocharger uses capscrews to attach the bearing housing to the turbine housing.

Position the turbocharger drain tube. The drain tube **must** be within 35 degrees of vertical. Turn the bearing housing to align the tube, if necessary.

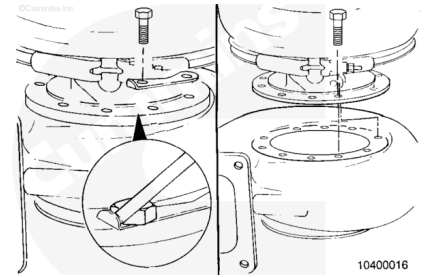


Bearing Housing Adjustment

Bend the lockplate off the capscrew heads.

Remove the capscrews to allow the bearing housing to rotate.

Lift the compressor and bearing housing. Rotate the bearing housing. Align the drain tube and the capscrew holes.



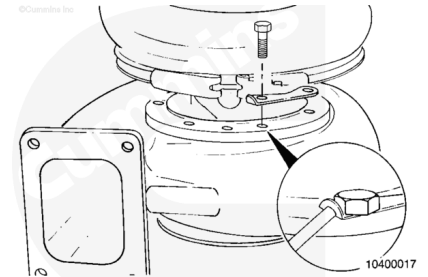
Apply anti-seize compound, Part Number 3824879, to the capscrews.

Install the lockplates and capscrews.

Tighten the capscrews.

Torque Value: 21 n•m [186 in-lb]

Bend the lockplate tabs over the capscrews.



High-Pressure Turbocharger:

The HE850 turbocharger does **not** use lockplate tabs.

If turbine housing orientation with respect to bearing housing does **not** match turbocharger being replaced, place turbocharger on turbine housing outlet flange on clean sturdy work bench.

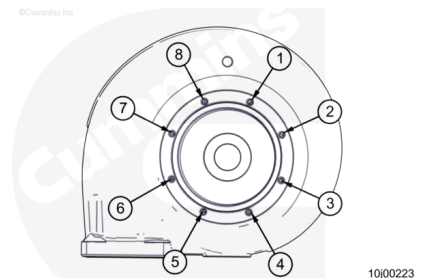
Secure the turbocharger and remove turbine housing to bearing housing mounting capscrews (3).

Rotate turbine housing to match orientation of turbocharger being replaced.

Install turbine housing to bearing housing mounting capscrews.

When torquing the turbine housing to bearing housing mounting capscrews, use the torque values and order of sequence shown below.

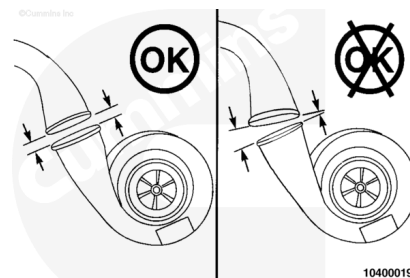
Turbine housing bolts shall be double tightened. Tighten bolts once in the circular sequence shown and repeat tightening each bolt again to the same value in the same circular sequence.



Torque Value: 40 n•m [30 ft-lb]

Check the air crossover-to-turbocharger alignment.

- Install the air crossovers on the aftercooler.
- Look at the alignment between the crossover and the turbocharger compressor outlet.
- Remove the crossover and proceed with the assembly if the alignment is correct.
- Proceed to the next step to adjust the compressor housing if the alignment is **not** correct.

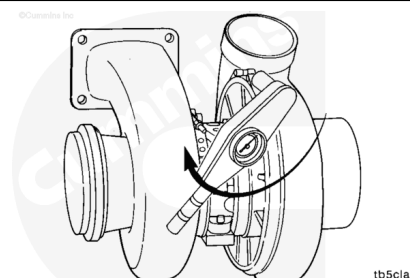


To adjust the turbocharger compressor housing alignment:

- Loosen the V-band clamp.
- Turn the compressor housing to the correct alignment.
- Tighten the clamp.

Torque Value: 9 n·m [80 in-lb]

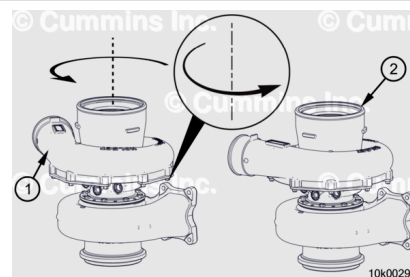
- Tap the clamp with a mallet. Tighten the clamp again.



High-Pressure Turbocharger:

⚠ CAUTION ⚠

Do not lift compressor housing. Axial movement of compressor housing can result in component damage.



⚠ CAUTION ⚠

Removing more than one clamp plate at a time can result in component damage.

The HE850 turbocharger uses a bolted compressor housing joint.

If compressor housing orientation with respect to diffuser does **not** match turbocharger being replaced (2), place turbocharger on turbine housing outlet flange on clean sturdy work bench.

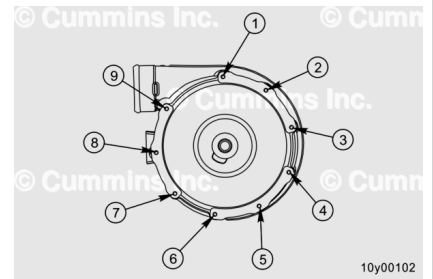
Secure the turbocharger and loosen compressor housing to diffuser mounting capscrews and clamp plates until housing rotates freely.

Do **not** lift compressor housing.

Rotate compressor housing to match orientation of turbocharger being replaced (1).

Tighten the capscrews in the order shown.

Compressor housing bolts shall be double tightened. Tighten bolts once in the circular sequence shown and repeat tightening each bolt again to the same value in the same circular sequence.



Torque Value:

M8 Bolts 33 n•m [24 ft-lb]

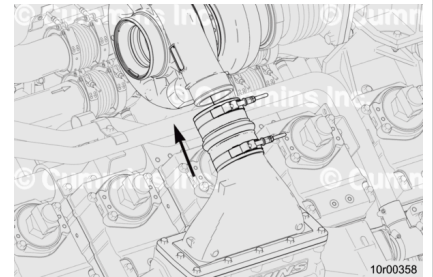
Torque Value:

M10 Bolts 50 n•m [37 ft-lb]

Install the hose and two hose clamps between each turbocharger and aftercooler inlet connection.

Tighten the clamps.

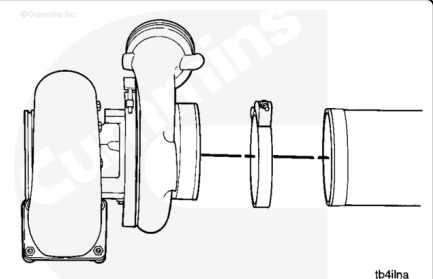
Torque Value: 9 n•m [80 in-lb]



Connect the inlet piping to the turbocharger.

Tighten the clamps.

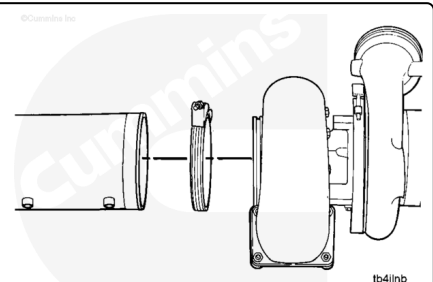
Torque Value: 9 n•m [80 in-lb]



Connect the exhaust piping to the turbocharger.

Tighten the V-band clamp.

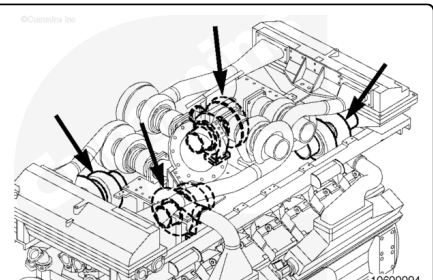
Torque Value: 9 n•m [80 in-lb]



QSK60 Two Stage Industrial and Power Generation

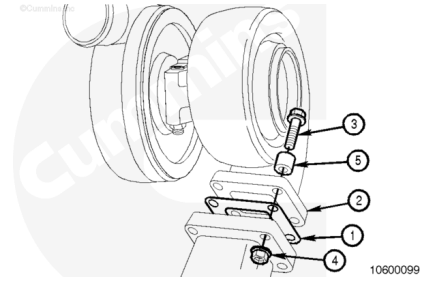
Determine which of the four high-pressure turbochargers requires installation:

- Left bank rear
- Right bank front
- Left bank front
- Right bank rear.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : Make sure that the turbocharger is correctly aligned during installation to prevent damage to the turbocharger bellows

Apply an anti-seize compound to the mounting capscrews threads.

The word OUT or the raised gasket bead **must** be positioned toward the turbocharger.

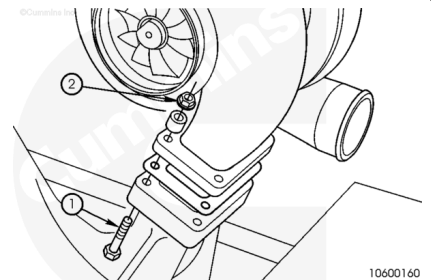
Install the gasket.

Do **not** install the turbocharger without spacers.

Install the turbocharger (2) and gasket (1) with four capscrews (3), spacers (5), and flange nuts (4).

⚠ CAUTION ⚠

Do not reuse old capscrews (1) or flange nuts (2). Discard these components. Reusing these components can contribute to exhaust leaks.

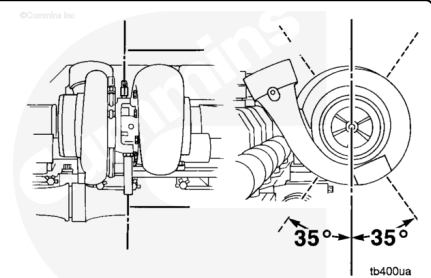


Tighten the flange nuts (4).

Torque Value: 55 n•m [40 ft-lb]

Position the turbocharger drain tube. The drain tube **must** be within 35 degrees of vertical. Turn the bearing housing to align the tube, if necessary.

The turbocharger uses capscrews to attach the bearing housing to the turbine housing.

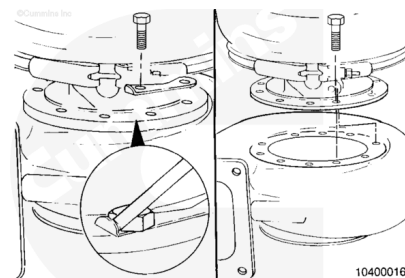


Bearing Housing Adjustment

Bend the lockplate off the capscrew heads.

Remove the capscrews.

Rotate the bearing housing. Align the drain tube and the capscrew holes.

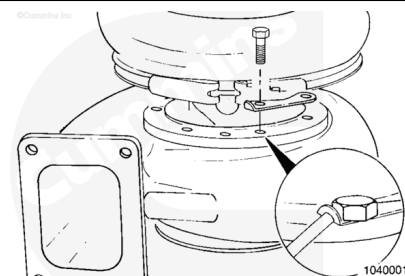


Install the lockplates and capscrews.

Tighten the capscrews.

Torque Value: 21 n•m [186 in-lb]

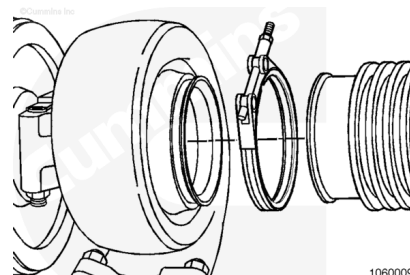
Bend the lockplate tabs over the capscrews.



Install the V-band clamp between the bellows and turbine side of the turbocharger.

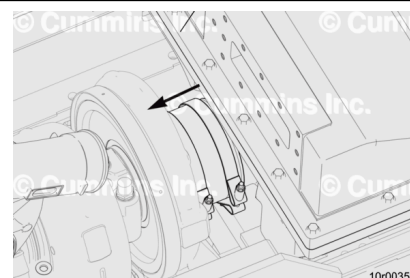
Tighten the V-band clamp.

Torque Value: 9 n•m [80 in-lb]



Install the hose and two hose clamps between the turbocharger inlet and the intercooler.

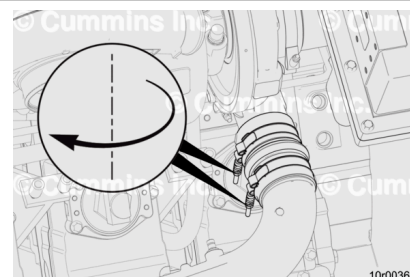
Slide the hose toward the turbocharger.



Install the hose and two hose clamps between each turbocharger and aftercooler inlet connection.

Tighten the clamps.

Torque Value: 9 n•m [80 in-lb]

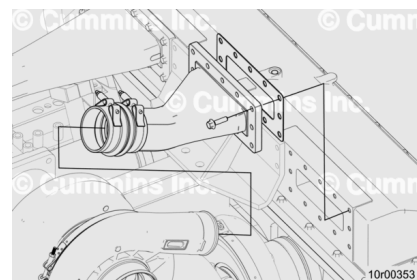


Left Bank Rear or Right Bank Front

Connect the intercooler intake air crossover, located between the low-pressure turbocharger and the intercooler, to the intercooler.

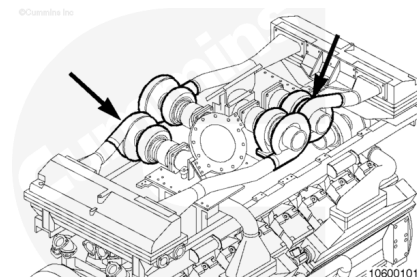
Tighten the 12 capscrews.

Torque Value: 45 n•m [33 ft-lb]



The left bank rear and right bank front turbochargers are located under the low-pressure turbochargers.

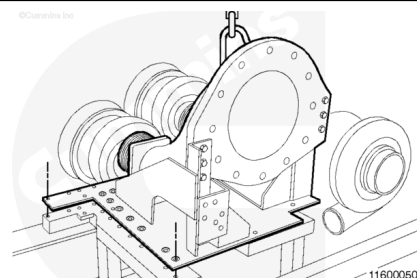
Install the low-pressure turbocharger directly above the high-pressure turbocharger being installed. Refer to Procedure 010-035 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-035-tr.html)



Left Bank Front or Right Bank Rear

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⚠ CAUTION ⚠

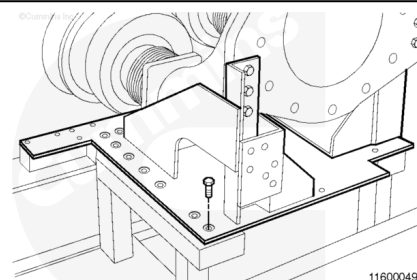
Care must be taken not to damage the bellows mounted to the exhaust outlet flange while setting the mounting plate in place.

Position the exhaust outlet flange and mounting base on the two-stage frame.

Install the thirty capscrews connecting the exhaust outlet mounting base to the two-stage frame.

Tighten the thirty mounting capscrews.

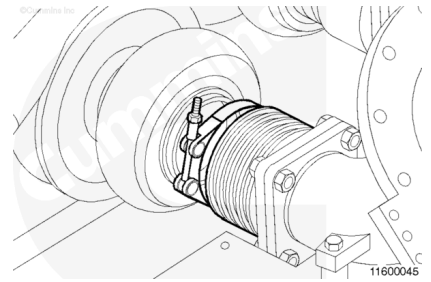
Torque Value: 45 n•m [33 ft-lb]



Connect the bellows on each leg of the exhaust outlet flange to the low-pressure turbocharger outlet.

Tighten the V-band clamps.

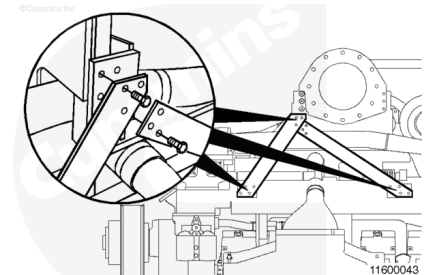
Torque Value: 8 n•m [71 in-lb]



Install the two support braces between the two-stage frame and the exhaust outlet connection mounting base.

Tighten the seven capscrews on each support brace.

Torque Value: 80 n•m [59 ft-lb]



Install the top cross support brace (1) between the two exhaust outlet connections (2).

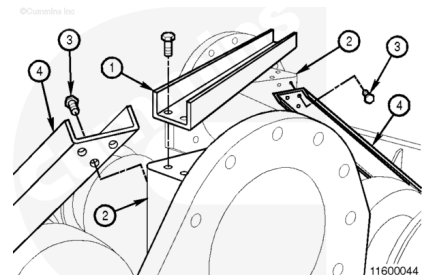
Tighten the eight capscrews (3).

Torque Value: 80 n•m [59 ft-lb]

Install the two inboard support struts (4).

Tighten the three capscrews (3) at each end.

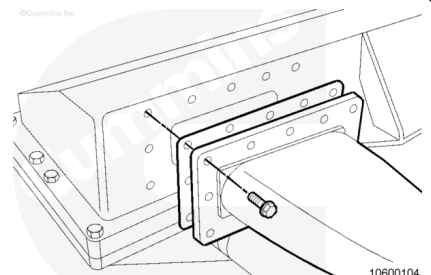
Torque Value: 45 n•m [33 ft-lb]



Connect the intercooler intake air crossover, located between the low-pressure turbocharger and the intercooler, to the intercooler.

Tighten the twelve capscrews.

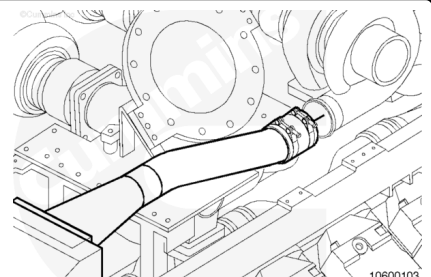
Torque Value: 45 n•m [33 ft-lb]



Install the hose clamp connecting the intercooler intake air crossover and the low-pressure turbocharger.

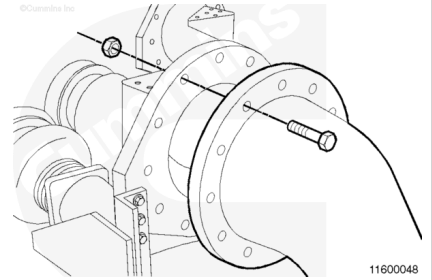
Tighten the hose clamps.

Torque Value: 8 n•m [71 in-lb]



Install the exhaust piping on the exhaust outlet flange with twelve capscrews and nuts.

Tighten the capscrews.



Last Modified: 18-Oct-2022
